WEB DEVELOPMENT

1. Introduction to Web Development

Q1. What is web development?

A) Designing hardware
B) Building websites
C) Writing OS
D) Networking

Q2. Frontend development deals with:

A) Database
B) Server
C) User interface
D) APIs

Q3. Backend handles:

A) Styling
B) Logic & database
C) Layout
D) Fonts

Q4. Which is frontend language?

A) Python
B) HTML
C) Java
D) SQL

Q5. Static website means:

A) Dynamic content
B) Fixed content

C) Database required
D) Server-side logic

Q6. Dynamic website uses:

A) Only HTML
B) Database
C) No server
D) CSS only

Q7. Full stack developer works on:

A) Only frontend
B) Only backend
C) Both
D) None

Q8. Browser is:

A) Server
B) Client software
C) Database
D) OS

Q9. Example of browser:

A) Apache
B) Chrome
C) MySQL
D) Linux

Q10. Server does:

A) Display UI
B) Stores files

C) Styling
D) Animations

Q11. URL stands for:

A) Uniform Resource Locator
B) Unique Resource Link
C) Universal Resource Loader
D) None

Q12. HTTPS is:

A) Unsecured
B) Encrypted
C) Slow
D) Old

Q13. Port of HTTP:

A) 443
B) 21
C) 80
D) 25

Q14. Domain name is:

A) IP address
B) Website name
C) Browser
D) Server

Q15. Hosting means:

A) Designing
B) Storing website

C) Coding
D) Testing

Q16. Git is used for:

A) Hosting
B) Version control
C) Styling
D) Database

Q17. HTML is used for:

A) Logic
B) Styling
C) Structure
D) Database

Q18. CSS is used for:

A) Structure
B) Styling
C) Logic
D) Backend

Q19. JavaScript is used for:

A) Styling
B) Logic/interactivity
C) Structure
D) Hosting

Q20. Example of web server:

A) Chrome
B) Apache

C) VS Code
D) Git

Q21. Which is dynamic site?

A) Portfolio
B) Blog with login
C) Resume
D) Static page

Q22. Full stack includes:

A) HTML only
B) Backend only
C) Frontend + Backend
D) Database only

Q23. Cloud hosting is:

A) Local storage
B) Online hosting
C) Offline hosting
D) USB storage

Q24. Browser sends request to:

A) Database
B) Server
C) CSS
D) HTML

Q25. Response comes from:

A) Browser
B) Server

C) User
D) CSS

2. Web Architecture & Protocols

Q1. Web architecture defines:
A) UI design
B) Interaction of components
C) Coding style
D) Database

Q2. Client-server model:
A) Server requests
B) Client requests
C) Both same
D) None

Q3. Client example:
A) Apache
B) Browser
C) MySQL
D) Linux

Q4. Server example:
A) Chrome
B) Apache
C) CSS
D) HTML

Q5. Protocol is:
A) Language
B) Rule of communication
C) Software
D) OS

Q6. HTTP stands for:

A) Hyper Transfer Text Protocol
B) HyperText Transfer Protocol
C) High Transfer Protocol
D) None

Q7. HTTPS provides:
A) Speed
B) Security
C) Storage
D) UI

Q8. HTTP is:
A) Stateful
B) Stateless
C) Secure
D) Database

Q9. GET method:
A) Sends data
B) Retrieves data
C) Deletes data
D) Updates data

Q10. POST method:
A) Retrieve
B) Send data
C) Delete
D) Update

Q11. PUT is used to:
A) Retrieve
B) Delete
C) Update
D) Store

Q12. DELETE method:

A) Update
B) Remove
C) Add
D) Retrieve

Q13. Status code 200:
A) Error
B) OK
C) Redirect
D) Fail

Q14. Status code 404:
A) Success
B) Not Found
C) Error
D) OK

Q15. DNS does:
A) Stores data
B) Converts domain to IP
C) Styling
D) Hosting

Q16. TCP ensures:
A) Speed
B) Reliability
C) UI
D) Design

Q17. IP handles:
A) Security
B) Addressing
C) Data
D) UI

Q18. OSI model has:

A) 5 layers
B) 6 layers
C) 7 layers
D) 8 layers

Q19. REST uses:
A) FTP
B) HTTP
C) SMTP
D) TCP only

Q20. API is:
A) UI
B) Communication interface
C) Database
D) Protocol

Q21. JSON is:
A) Language
B) Data format
C) Server
D) Protocol

Q22. Cookies are stored in:
A) Server
B) Browser
C) Database
D) Cloud

Q23. Sessions stored in:
A) Browser
B) Server
C) Client
D) Cache

Q24. Stateless means:

A) Memory retained
B) No memory
C) Secure
D) Fast

Q25. First step in request flow:
A) HTTP request
B) DNS lookup
C) Rendering
D) Server response

3. HTML Fundamentals

Q1. HTML stands for:

A) Hyper Tool Markup Language
B) HyperText Markup Language
C) HighText Machine Language
D) None

Q2. HTML is:

A) Programming language
B) Markup language
C) Database
D) OS

Q3. Root element:

A) head
B) body
C) html
D) title

Q4. Visible content inside:

A) head
B) body
C) title
D) html

Q5. Tag example:

A) p
B) <p>
C) paragraph
D) text

Q6. Attribute example:

A) p
B) src
C) tag
D) html

Q7. Largest heading:

A) h6
B) h3
C) h1
D) h2

Q8. Paragraph tag:

A) <para>
B) <p>
C) <text>
D) <pg>

Q9. Bold tag:

A) <b>
B) <bold>
C) <stronger>
D) <bd>

Q10. Link tag:

A) <link>
B) <a>
C) <href>
D) <url>

Q11. Image tag:

A) <img>
B) <image>
C) <pic>
D) <src>

Q12. Table row:

A) <td>
B) <tr>
C) <th>
D) <row>

Q13. Table data:

A) <tr>
B) <td>
C) <th>
D) <table>

Q14. Form tag:

A) <input>
B) <form>
C) <button>
D) <submit>

Q15. Input type password hides:

A) Name
B) Text
C) Password
D) Email

Q16. Semantic tag:

A) div
B) span
C) header
D) b

Q17. Block element:

A) span
B) a
C) div
D) img

Q18. Inline element:

A) div
B) p
C) span
D) section

Q19. Comment syntax:

A) //
B) <!-- -->
C) ##
D) **

Q20. Audio tag:

A) <music>
B) <sound>
C) <audio>
D) <mp3>

Q21. Video tag:

A) <vid>
B) <video>
C) <media>
D) <movie>

Q22. Ordered list:

A) ul
B) li
C) ol
D) list

Q23. Unordered list:

A) ol
B) ul
C) li
D) list

Q24. HTML5 introduced:

A) tables
B) semantic tags
C) lists
D) links

Q25. Span is:

A) Block
B) Inline
C) Table
D) Form

CSS Fundamentals

1. CSS stands for:

A) Color Style Sheets
B) Cascading Style Sheets
C) Creative Style Sheets
D) None

Q2. CSS is used for:

A) Logic
B) Styling
C) Database
D) Server

Q3. Best CSS type:

A) Inline
B) Internal
C) External
D) None

Q4. Selector defines:

A) Value
B) Style target
C) Property
D) Color

Q5. Class selector uses:

A) #
B) .
C) *
D) @

Q6. ID selector uses:

A) .
B) #
C) *
D) $

Q7. Universal selector:

A) .
B) #
C) *
D) all

Q8. Property example:

A) red
B) color
C) h1
D) class

Q9. Value example:

A) color
B) red
C) p
D) id

Q10. HEX color:

A) rgb()
B) red
C) #ff0000
D) color

Q11. Box model includes:

A) Padding
B) Margin
C) Border
D) All

Q12. Padding is:

A) Outside
B) Inside
C) Border
D) Height

Q13. Margin is:

A) Inside
B) Outside
C) Border
D) Content

Q14. Display block:

A) Inline
B) Full width
C) Hidden
D) None

Q15. Inline element:

A) Full width
B) Limited width
C) Block
D) Fixed

Q16. Position default:

A) Relative
B) Static
C) Fixed
D) Absolute

Q17. Flexbox is used for:

A) Colors
B) Layout
C) Fonts
D) Borders

Q18. Opacity:

A) Size
B) Transparency
C) Color
D) Border

Q19. Z-index:

A) Color
B) Stack order
C) Size
D) Margin

Q20. Border syntax:

A) border: red
B) border: 2px solid red
C) border: solid
D) border: px

Q21. Background property:

A) text
B) color
C) background-color
D) font

Q22. Overflow hidden:

A) Show all
B) Hide extra
C) Scroll
D) Resize

Q23. External CSS file:

A) .html
B) .css
C) .js
D) .txt

Q24. Flex container:

A) display: block
B) display: flex
C) display: inline
D) display: none

Q25. Inline CSS:

A) Inside file
B) In tag
C) External
D) Server

Advanced CSS & Responsive Design

Q1. Responsive design means:

A) Fixed layout
B) Adaptive layout
C) Static page
D) None

Q2. Viewport meta tag is used for:

A) Styling
B) Mobile scaling
C) Animation
D) Fonts

Q3. Media query applies:

A) Always
B) Conditionally
C) Never
D) Random

Q4. Flexbox is:

A) 2D layout
B) 1D layout
C) Grid
D) Table

Q5. Grid is:

A) 1D
B) 2D
C) Inline
D) Block

Q6. Flex direction default:

A) Column
B) Row
C) Grid
D) None

Q7. justify-content aligns:

A) Vertical
B) Horizontal
C) Both
D) None

Q8. align-items aligns:

A) Horizontal
B) Vertical
C) Both
D) None

Q9. display flex:

A) Inline
B) Block
C) Flex container
D) None

Q10. display grid:

A) Table
B) Layout system
C) Inline
D) None

Q11. Media query syntax:

A) @style
B) @media
C) @css
D) @query

Q12. max-width means:

A) Minimum
B) Maximum
C) Fixed
D) None

Q13. Sticky position:

A) Fixed always
B) Scroll + fixed
C) Static
D) Absolute

Q14. Transition is:

A) Instant
B) Smooth change
C) Layout
D) Grid

Q15. Hover effect:

A) Click
B) Mouse over
C) Scroll
D) Load

Q16. Animation uses:

A) @animate
B) @keyframes
C) @move
D) @run

Q17. transform scale:

A) Rotate
B) Resize
C) Move
D) Hide

Q18. vw unit:

A) Pixel
B) View width
C) Font
D) Height

Q19. vh unit:

A) Width
B) Height
C) Font
D) Margin

Q20. Mobile-first:

A) Desktop first
B) Mobile priority
C) Tablet only
D) None

Q21. Breakpoints define:

A) Colors
B) Screen sizes
C) Fonts
D) Margin

Q22. Grid columns:

A) grid-col
B) grid-template-columns
C) col-grid
D) template

Q23. Gap property:

A) Margin
B) Space between items
C) Padding
D) Border

Q24. Flexbox best for:

A) Tables
B) Alignment
C) Database
D) Images

Q25. Grid best for:

A) Alignment
B) Complex layouts
C) Text
D) Animation

JavaScript Fundamentals/Advanced JavaScript

Q1. JavaScript is:

A) Markup
B) Programming
C) Styling
D) Database

Q2. Correct variable:

A) let x
B) let x = 5
C) x let = 5
D) var = 5

Q3. const means:

A) Changeable
B) Fixed
C) Loop
D) Function

Q4. Data type:

A) string
B) number
C) boolean
D) All

Q5. === checks:

A) Value
B) Type
C) Both
D) None

Q6. Function keyword:

A) func
B) function
C) def
D) fn

Q7. Arrow function uses:

A) ->
B) =>
C) ==
D) <=

Q8. Array:

A) {}
B) []
C) ()
D) <>

Q9. Object:

A) []
B) {}
C) ()
D) ""

Q10. DOM is:

A) Data
B) Structure of page
C) Style
D) Server

Q11. Event click:

A) hover
B) action
C) load
D) error

Q12. Scope defines:

A) Data
B) Accessibility
C) Loop
D) Event

Q13. Hoisting:

A) Down
B) Up
C) Side
D) None

Q14. Closure:

A) Loop
B) Inner function memory
C) Object
D) Array

Q15. Promise:

A) Sync
B) Async
C) Loop
D) Variable

Q16. Async keyword:

A) Makes sync
B) Makes async
C) Stops code
D) None

Q17. await works with:

A) Loop
B) Promise
C) Function
D) Object

Q18. Fetch is:

A) API call
B) Loop
C) Object
D) Event

Q19. Spread operator:

A) ...
B) ***
C) ###
D) ---

Q20. Destructuring:

A) Merge
B) Extract
C) Delete
D) Loop

Q21. Callback:

A) Return
B) Function passed
C) Loop
D) Object

Q22. Event loop handles:

A) Sync
B) Async
C) Variables
D) CSS

Q23. setTimeout:

A) Delay
B) Loop
C) Condition
D) Function

Q24. localStorage stores:

A) Temp
B) Permanent
C) Session
D) None

Q25. try block:

A) Error
B) Code
C) Loop
D) Data

Q26. catch block:

A) Handles error
B) Loop
C) Function
D) Variable

Q27. let scope:

A) Global
B) Block
C) None
D) Function

Q28. var scope:

A) Block
B) Function
C) Global only
D) None

Q29. Promise states:

A) Pending
B) Resolved
C) Rejected
D) All

Q30. Async code runs:

A) Blocking
B) Non-blocking
C) Sync
D) None

REMAINING TOPICS

FRONTEND + BACKEND (1–20)

1. What is React?
   A. Framework
   B. Library
   C. Language
   D. Database
   Answer: B

2. Angular is a:
   A. Library
   B. Framework
   C. Database
   D. Language
   Answer: B

3. Virtual DOM improves:
   A. Security
   B. Speed
   C. Storage
   D. SEO
   Answer: B

4. Props are:
   A. Internal state
   B. External data
   C. Database
   D. API
   Answer: B

5. MVC stands for:
   A. Model View Controller
   B. Main View Control
   C. Model Value Code
   D. None
   Answer: A

6. Backend handles:
   A. UI
   B. Styling
   C. Logic
   D. Animation
   Answer: C

7. Node.js uses:
   A. Python
   B. JavaScript
   C. Java
   D. PHP
   Answer: B

8. Middleware is:
   A. Database
   B. Function between request-response
   C. UI
   D. API
   Answer: B

9. Server responds using:
   A. CSS
   B. HTTP
   C. HTML only
   D. XML
   Answer: B

10. Express.js is:
    A. Database
    B. Backend framework
    C. Browser
    D. OS
    Answer: B

DATABASES (21–35)

21. SQL stands for:
    A. Structured Query Language

B. Simple Query Language
C. Server Query Logic
D. None
Answer: A

22. MongoDB is:
A. SQL
B. NoSQL
C. API
D. Server
Answer: B

23. CRUD means:
A. Create Read Update Delete
B. Code Run Update Delete
C. Create Remove Use Data
D. None
Answer: A

24. Primary key is:
A. Unique identifier
B. Duplicate value
C. Foreign key
D. Index
Answer: A

25. Indexing improves:
A. Speed
B. Security
C. UI
D. Storage
Answer: A

26. ORM stands for:
A. Object Relational Mapping
B. Object Resource Model
C. Online Resource Manager
D. None
Answer: A

27. Foreign key links:
    A. Tables
    B. Servers
    C. APIs
    D. UI
    Answer: A

28. NoSQL stores:
    A. Tables only
    B. Documents
    C. Rows only
    D. Keys only
    Answer: B

29. PostgreSQL is:
    A. NoSQL
    B. SQL
    C. API
    D. Framework
    Answer: B

30. Many-to-many needs:
    A. One table
    B. Two tables
    C. Junction table
    D. No table
    Answer: C

SECURITY (36–55)

36. XSS attacks:
    A. Server
    B. Browser
    C. Database
    D. Network
    Answer: B

37. SQL Injection targets:
    A. CSS

B. Database
C. UI
D. API
Answer: B

38. CSRF stands for:
A. Cross Site Request Forgery
B. Code Safe Request
C. Client Server Response
D. None
Answer: A

39. HTTPS uses:
A. SSL/TLS
B. HTML
C. CSS
D. JS
Answer: A

40. Hashing is used for:
A. UI
B. Password security
C. API
D. CSS
Answer: B

41. bcrypt is:
A. Database
B. Hashing algorithm
C. API
D. Framework
Answer: B

42. DDoS attacks:
A. Database
B. Server overload
C. UI
D. CSS
Answer: B

43. Authentication checks:
    A. Permissions
    B. Identity
    C. UI
    D. CSS
    Answer: B

44. Authorization checks:
    A. Identity
    B. Permissions
    C. UI
    D. Server
    Answer: B

45. Input validation prevents:
    A. UI bugs
    B. Attacks
    C. CSS errors
    D. None
    Answer: B

PERFORMANCE (56–70)

56. Minification removes:
    A. Bugs
    B. Spaces
    C. Data
    D. APIs
    Answer: B

57. CDN stands for:
    A. Content Delivery Network
    B. Code Delivery Node
    C. Content Data Node
    D. None
    Answer: A

58. Lazy loading loads:
    A. All data

B. Needed data
C. No data
D. API
Answer: B

59. Caching improves:
A. Speed
B. Security
C. UI
D. Storage
Answer: A

60. LCP measures:
A. Speed
B. UI
C. Data
D. API
Answer: A

TESTING (71–80)

71. Unit testing tests:
A. Whole system
B. Small units
C. UI
D. API
Answer: B

72. Integration testing checks:
A. Modules together
B. UI
C. CSS
D. Server
Answer: A

73. E2E testing simulates:
A. Database
B. User
C. API

D. Server
Answer: B

74. Jest is:
A. Database
B. Testing tool
C. API
D. UI
Answer: B

75. Selenium is used for:
A. Backend
B. Automation
C. Database
D. API
Answer: B

DEPLOYMENT + APIs + MODERN DEV (81–100)

81. Deployment means:
A. Coding
B. Production release
C. Testing
D. Debugging
Answer: B

82. AWS is:
A. Cloud
B. Database
C. API
D. UI
Answer: A

83. CI stands for:
A. Continuous Integration
B. Code Integration
C. Control Input
D. None
Answer: A

84. REST API uses:
    A. HTTP
    B. CSS
    C. HTML
    D. JS
    Answer: A

85. GET request:
    A. Sends data
    B. Fetches data
    C. Deletes data
    D. Updates
    Answer: B

86. POST request:
    A. Fetch
    B. Send
    C. Delete
    D. Update
    Answer: B

87. JSON is:
    A. Data format
    B. Language
    C. DB
    D. API
    Answer: A

88. GraphQL:
    A. Fixed data
    B. Flexible queries
    C. No API
    D. CSS
    Answer: B

89. SPA means:
    A. Single Page App
    B. Server Page App
    C. Simple Page App

D. None
Answer: A

90. Git is:
A. Version control
B. Database
C. API
D. UI
Answer: A

91. Microservices:
A. Monolith
B. Small services
C. UI
D. CSS
Answer: B

92. Serverless means:
A. No backend
B. Managed servers
C. No API
D. No DB
Answer: B

93. JAMstack includes:
A. JS APIs Markup
B. Java API Model
C. JSON API Model
D. None
Answer: A

94. Agile is:
A. Methodology
B. Language
C. API
D. DB
Answer: A

95. Code review improves:
A. UI
B. Quality

C. Speed
D. API
Answer: B

96. Endpoint is:
A. API URL
B. DB
C. UI
D. CSS
Answer: A

97. PUT request:
A. Create
B. Update
C. Fetch
D. Delete
Answer: B

98. DELETE request:
A. Remove
B. Fetch
C. Send
D. Update
Answer: A

99. Debugging finds:
A. Features
B. Bugs
C. APIs
D. DB
Answer: B

100. DevTools are used for:
A. Coding
B. Debugging
C. Hosting
D. DB
Answer: B